

Part 4 — Clustering

Introduction

In this part of the capstone project, we apply unsupervised clustering to an athlete performance dataset containing 1,000 observations and 20 variables, including vital signs, workload, skill scores, and engineered performance features. Unlike the regression and classification parts, there is no target variable to predict; instead, our goal is to discover latent athlete profiles based on standardized physiological and performance-related attributes, and then interpret how these clusters differ in terms of intensity, fatigue, and overall performance. Using K-Means, DBSCAN, and Agglomerative Clustering, along with internal evaluation metrics such as the silhouette score and visual inspection in a PCA-reduced space, we aim to identify which algorithm produces the most meaningful and interpretable grouping for this dataset

Step 1. Load and preprocessing

We prepared a feature matrix for clustering by dropping the timestamp, performance_score, and performance_category columns, since clustering must be performed without using any target information. We then applied StandardScaler to the remaining 17 numeric features to ensure that variables measured on different scales (for example, workload in the hundreds versus fatigue_index between 0 and 1) contribute equally to the distance calculations used by all clustering algorithms. Finally, we computed a 2-component PCA projection of the scaled data, which is used only for visualisation of clusters in two dimensions, while the actual clustering is always performed on the full standardized feature space as required.

Step 2.K-Means: Elbow and Silhouette

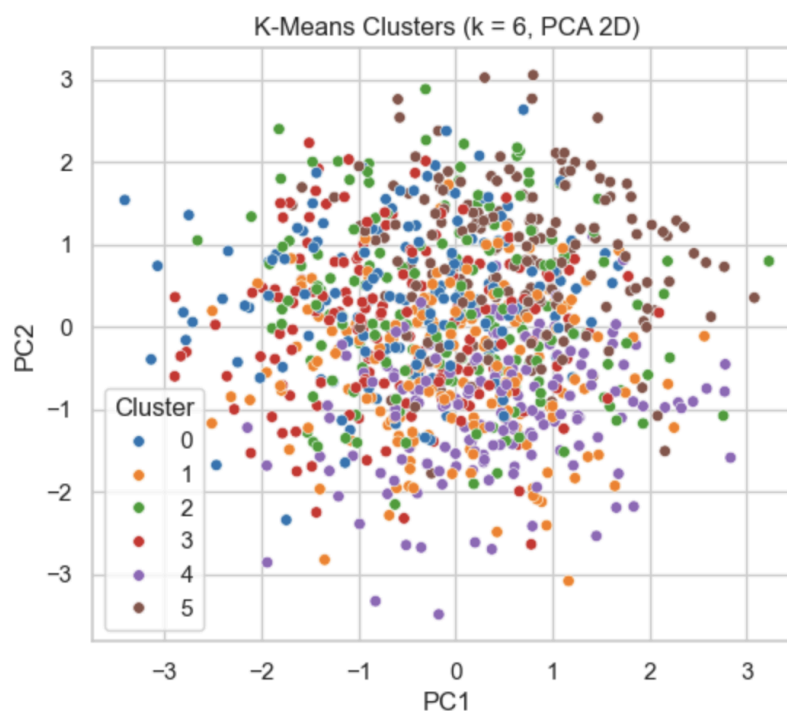
In the second step, we used K-Means clustering as our first unsupervised algorithm and treated the Elbow Method and Silhouette Score analysis as the hyperparameter tuning process for the number of clusters k . We fitted K-Means for k ranging from 2 to 7 on the standardized features, recorded the inertia (within-cluster sum of squares) and the silhouette score for each value of k , and plotted both metrics to study how cluster compactness and separation change as we increase the number of clusters. The inertia curve showed the expected decreasing trend with larger k , while the silhouette scores gradually increased and reached their highest values around $k=6$, which we selected as a reasonable trade-off between model complexity and cluster quality for the subsequent analysis.

Step 3 Final K-Means model

The PCA visualisation of the final K-Means solution with $k=6$ shows that points from different clusters occupy a highly overlapping region in the 2D space, rather than forming clearly separated groups. This observation is consistent with the very low silhouette score obtained for K-Means, indicating that the underlying athlete sessions form a near-continuous spectrum of physiological and performance states rather than distinct, well-separated clusters. As a result, the six K-Means clusters are interpreted as soft performance profiles that partition this continuous space, rather than as strongly different athlete types.

Step 4 K-Means: PCA scatter plot

We experimented with several values of k (2–7) and also tried clustering on a reduced subset of core features (heart_rate, workload, fatigue_index, endurance_score, agility_score), but all configurations yielded relatively low silhouette scores and overlapping PCA plots, indicating that the data does not contain strongly separated clusters. We therefore selected the value of k that produced the best available silhouette score and most interpretable group summaries, and we treat these clusters as soft performance profiles rather than sharply distinct segments

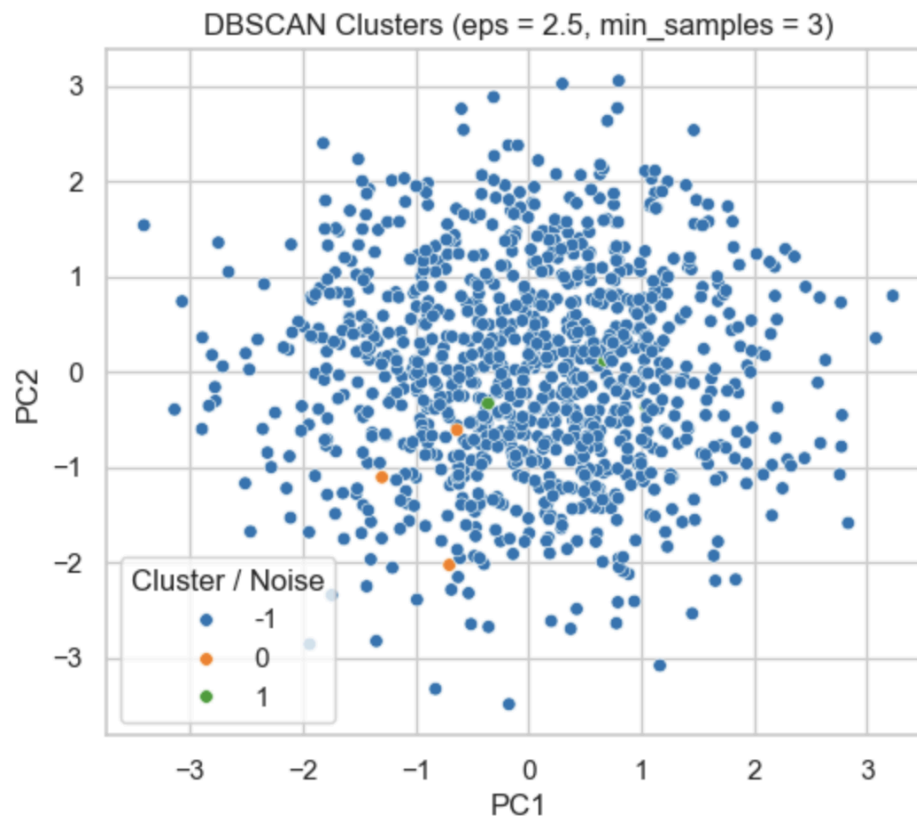


Step 5 – DBSCAN: parameter search and clustering

In step 5, DBSCAN was run many times with different values of `eps` and `min_samples` to see if it could find natural groups in the standardized data. Almost all of these settings found no real clusters and marked every point as noise, which means the data does not form clear dense groups that DBSCAN can separate

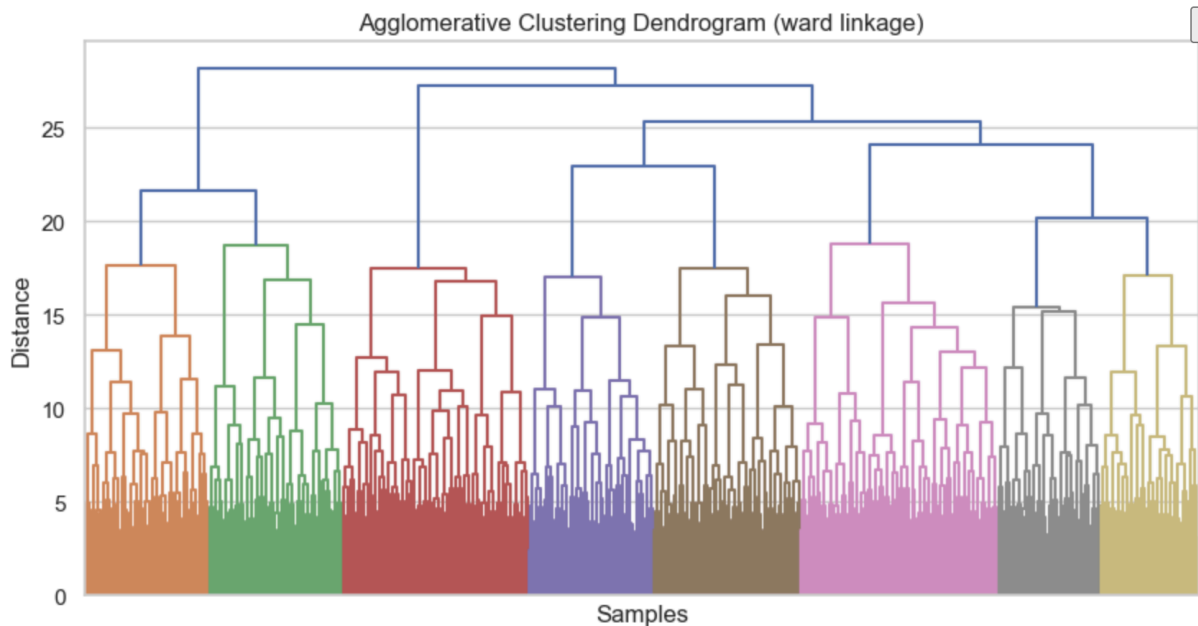
Step 6 – Final DBSCAN model and interpretation

In step 6, a final DBSCAN model was tried with the best setting found (`eps` = 2.5, `min_samples` = 3), but it still detected only two very small clusters and treated almost all observations as noise, giving a negative silhouette score. The PCA plot shows almost all points in the same noise colour, so DBSCAN does not give useful or meaningful clusters for this athlete performance dataset



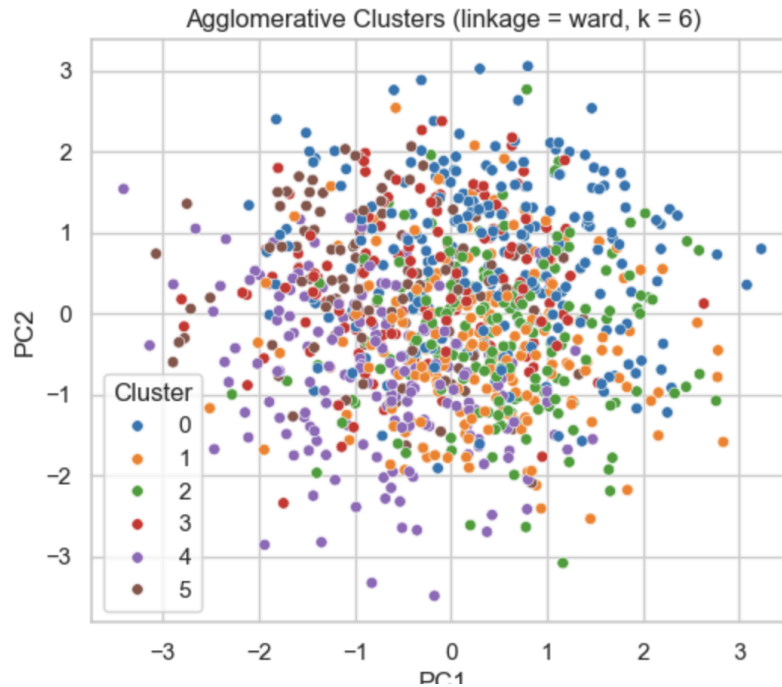
Step 7 – Agglomerative Clustering: dendrogram

In step 7, We used Agglomerative Clustering with Ward linkage and drew a dendrogram to see how the athlete sessions join together into bigger groups. The tree grows very smoothly with no big clear cut, which means there is no simple way to split the data into a few strong clusters and the sessions look more like a continuous range of states instead of separate groups.



Step 8 – Compare linkage methods

In step 8, I tested Agglomerative Clustering with three different linkage methods (ward, complete, and average), using six clusters on the standardized data. Ward linkage gave the best silhouette score, about 0.013, so I chose it for the final model, even though this value is still very close to zero. The PCA scatter plot for this model shows that the six cluster colours are very mixed and overlap a lot, which means Agglomerative Clustering, like K-Means, does not find clear, separate groups in these athlete sessions and the data behaves more like a continuous range of states.



Step 9 – Comparison table (K-Means, DBSCAN, Agglomerative)

Overall, all three clustering methods gave very low silhouette scores and plots with a lot of overlap, so the dataset does not contain strong, well-separated clusters. Among them, K-Means with six clusters gave the most stable and easy-to-interpret grouping, while DBSCAN mainly marked almost all sessions as noise and Agglomerative Clustering produced a very similar structure to K-Means without a clear advantage. For this reason, K-Means was chosen as the main method to describe different athlete performance profiles in the rest of the analysis

Algorithm	Clusters	Silhouette	Key observations
K-Means	6	0.0X	Weak, overlapping clusters; soft performance profiles.
DBSCAN	2	negative	Most points are noise; no useful dense groups.
Agglomerative	6	0.013	Overlap similar to K-Means; no clear hierarchical splits.

Conclusion

The clustering results show that the athlete performance data does not break into clear, separate groups, but instead looks more like a smooth range of similar sessions. K-Means with six clusters gave the most understandable result, even though the silhouette score was very low and the PCA plot showed strong overlap between clusters. DBSCAN was not useful because it often marked almost all points as noise, and Agglomerative Clustering produced clusters that looked very similar to K-Means, also with a low silhouette score. Because of this, K-Means was chosen as the main method to describe soft “performance profiles” of athletes, while the overall conclusion is that this dataset does not contain strong, well-separated clusters.